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R&D Project Proposal

Interactive Reinforcement Learning with Human Feedback Collected through VR Headsets

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1 Introduction

1.1 Topic of This R&D Project

Robots have the potential to become important partners in human environments, helping with tasks at home or in complex collaborative work [1, 5]. However, it is still very difficult for robots to work well in open-world environments. These environments have unknown tasks, new objects, and dynamic human interactions. This gap highlights an important shortcoming: robots often lack the common sense and contextual understanding that humans use effortlessly to generalize their skills and interpret underspecified instructions.

A promising way to address this problem is Interactive Reinforcement Learning (IRL), which includes human feedback directly into the robot’s learning loop [4, 16]. This approach aims to improve sample efficiency, accelerate policy acquisition, and align the robot’s behavior with human preferences and intentions [16]. By learning from evaluative feedback, demonstrations, or advice, IRL agents can adapt to complex, real-world constraints more effectively than autonomous reinforcement learning agents relying only on engineered reward functions.

Despite its potential, a significant bottleneck in traditional IRL is the interface for human feedback. Most systems rely on low-bandwidth, explicit channels such as keyboard presses, button clicks, or simple GUI interactions [11, 18]. These methods are unnatural and awkward for the human teacher. They also cannot capture the rich details of human communication, like gestures, gaze, and tone of voice [17, 27]. This creates an expressiveness gap, limiting the quality and type of information that can be transferred from human to robot.

At the same time, advances in consumer-grade Virtual Reality (VR) technology have created new opportunities for human-computer interaction. Modern VR headsets have many sensors that can track head pose, hand gestures, eye gaze, and voice input, offering an immersive and intuitive interface for human communication. These sensors can capture detailed human feedback in a natural way [19, 28]. Even with this potential, VR and IRL have mostly developed separately. VR is often used to collect demonstration data (e.g., [9, 23]) and a recent study used it for preference-based learning [8]. However, using VR to collect real-time scalar

feedback as part of an IRL system remains largely unexplored.

This project aims to close this gap by developing a novel IRL system that uses VR as an interface for human feedback. The core idea is to allow users to interact with a virtual agent using gestures, gaze, voice, or controller inputs to deliver reinforcement signals in real time. The main goal of this research is to build a proof-of-concept system that demonstrates the feasibility of using immersive technologies for human-in-the-loop RL, making AI training more natural and intuitive.

1.2 Relevance of This R&D Project

This project is important for several reasons:

Scientific Value: This work explores the integration of rich, multi-modal feedback from VR headsets—such as pose, gaze direction, and hand gestures—into the IRL feedback loop. It goes beyond simple binary or scalar inputs [11, 18] and investigates the potential for more natural human-robot communication.

Practical Impact: The outcomes of this project will benefit applications that require efficient and intuitive human-in-the-loop training. In robotics, it can provide more natural training interfaces. Domain experts—such as rehabilitation therapists or simulation designers—could potentially train agents more naturally without requiring programming expertise. In the simulation and digital twin industry, animators and designers could use this system to create more realistic and responsive virtual characters [19, 28] with less manual effort. Domain experts—such as rehabilitation therapists or simulation designers—will be able to train agents more naturally and effectively, without requiring programming expertise.

Technological Advancement: The project will deliver a functional framework that connects popular VR development platforms (e.g., Unity engine) with standard RL libraries (e.g., PyTorch, ML-Agents). This provides useful infrastructure that allows other researchers to build upon our work and encourage collaboration in the community. We aim to lower the barrier to entry for future studies in immersive human-agent interaction.

2 Related Work

2.1 Reinforcement Learning

Reinforcement Learning (RL) is a machine learning paradigm where an autonomous agent learns to make sequential decisions by interacting with an environment. The core objective is to learn a policy—a mapping from states of the environment to actions—that maximizes a numerical reward signal over time [16]. This interaction is formally modeled as a Markov Decision Process (MDP), defined by the tuple (S, A, P, R, γ) , where:

- S is a set of states,
- A is a set of actions,
- $P(s_{t+1}|s_t, a_t)$ is the state transition probability,
- $R(s_t, a_t, s_{t+1})$ is the reward function, and
- $\gamma \in [0, 1]$ is a discount factor that weights the importance of immediate versus future rewards.

The agent’s goal is to learn an optimal policy $\pi^*(a|s)$ that maximizes the expected cumulative discounted reward, or return. A fundamental concept is the value function $V^\pi(s)$, which estimates the expected return from state s when following policy π , and the action-value function $Q^\pi(s, a)$, which estimates the expected return after taking action a in state s and thereafter following policy π .

A review of current research shows that IRL uses many different algorithms. Researchers choose these methods strategically based on the task and the type of human feedback available.

The algorithms used in human-robot interaction (HRI) can be divided into three main groups, each with different strengths:

2.1.1 Value-Based Methods: Stability and Integration

Value-based methods, especially Q-learning and its variants, are the most common type found in the literature. They are popular because they are well-understood, stable in discrete action spaces, and it is straightforward to add human feedback to them [16]. The foundational tabular Q-learning algorithm is effectively employed in

simpler, discrete state-action tasks, such as the sequential object selection task in Trick et al. [27] and the grid-world navigation and trust optimization in Alzahrani and Ahmad [2]. Its simplicity makes the impact of human advice or reward shaping directly observable. Furthermore, Munguia-Galeano et al. [24] uses a Contextual Q-learning (CQL) approach to leverage affordances, demonstrating how classic value-based methods can be extended for more complex, context-aware HRI. For higher-dimensional state spaces, such as pixel inputs in simulated environments, Deep Q-Networks represent the natural evolution. For instance, Moreira et al. [22] uses a DQN with a CNN for an object-sorting task, while Xu et al. [29] develops a more advanced Bayesian Deep Q-Network (BDQN) to robustly integrate the uncertainty inherent in EEG-based implicit feedback. This illustrates a trend where value-based methods are scaled using deep learning while maintaining their core principle of learning an action-value function.

2.1.2 Policy Search and Policy Gradient Methods: Direct Optimization

Policy Search methods directly optimize the policy parameters. This approach is often chosen for continuous control tasks common in robotics, like manipulation and locomotion. The Policy Learning by Weighting Exploration with Returns (PoWER) algorithm is a key part of policy search in robotics. Pahič et al. [25] uses PoWER in a latent space to learn a ball-throwing skill, demonstrating how policy search can be effectively guided by discrete human ratings instead of engineered rewards. Policy Gradient and Cross-Entropy Method are foundational policy optimization techniques. Sridharan [26] uses Policy Gradient and Cross-Entropy methods in their "Augmented RL" framework, showcasing their utility for online weight adaptation between human and environmental rewards. This highlights a key strength: the direct coupling between human feedback (which often critiques the policy) and the policy update rule.

2.1.3 Actor-Critic and Advanced Methods: Complexity and Continuous Control

Actor-Critic methods, which combine value function and policy learning, address complex, continuous control problems. Their ability to handle high-dimensional

action spaces makes them suitable for modern robotic simulation. Proximal Policy Optimization (PPO) is a modern actor-critic algorithm known for its stability, used in several advanced physics-based simulation studies. It is used for human motion tracking and avatar control in Winkler et al. [28], Ye et al. [30], and Luo et al. [19]. In these works, PPO is not typically interactive with a human during training but is used in an imitation learning setting, often trained on motion capture (MoCap) data, which can be seen as a form of offline human demonstration. Soft Actor-Critic (SAC) is an algorithm that maximizes both expected return and entropy (promoting exploration), used by Gonzalez-Santocildes et al. [12] for its sample efficiency and robustness in learning comfort-aware robot behaviors. Its off-policy nature makes it well-suited for learning from the potentially expensive data gathered in HRI experiments.

2.1.4 Human-Feedback-Specific Frameworks

A large part of the literature does not use a standard RL algorithm as its core. Instead, it builds a dedicated framework around human input. Introduced by Bradley Knox and Stone [4], TAMER (Training an Agent Manually via Evaluative Reinforcement) is a key framework that uses supervised learning to model a human’s reward function from scalar feedback. It is not RL in the traditional sense but is a cornerstone of IRL. Its influence is clear in studies like Lin et al. [17], which adapts its principles to interpret social feedback, and Faulkner et al. [11], which combines it with RL. Furthermore, several studies, such as Faulkner et al. [11] and Kessler Faulkner and Thomaz [13], propose meta-frameworks for Policy Shaping that are algorithm-agnostic, like REPaIR and AMPS. These are designed to filter or interpret human feedback and can be layered on top of various underlying RL algorithms (often Q-learning or policy gradients), making them flexible tools for handling imperfect teachers.

A primary challenge in RL is the sample inefficiency in learning from sparse environmental rewards. This project addresses this issue by integrating human feedback to accelerate and guide the learning process, moving beyond traditional RL frameworks that rely only on manually designed reward functions.

2.2 Survey of Related Work

Our work sits at the intersection of IRL, HRI, and VR interfaces. We review key contributions across these domains.

2.2.1 Interactive Reinforcement Learning (IRL)

The foundation of this review is based on important papers that shaped the IRL field. Key work by Bradley Knox and Stone [4] introduced TAMER, showing that human evaluative feedback could train agents much faster than autonomous RL. This was extended into social domains by Lin et al. [17] and Lin et al. [16] who provide a comprehensive review of IRL from human social feedback, distinguishing between evaluative feedback (e.g., rewards or punishments that signal approval or disapproval) and instructive feedback (e.g., advice, corrections, or demonstrations that guide the learner toward desired behaviors).

A number of IRL frameworks have been proposed to integrate human feedback in different ways. COACH uses human feedback directly to calculate the policy gradient in an actor-critic algorithm. This allows for more flexible guidance, particularly in continuous action spaces, and supports real-time corrective feedback during learning [16].

Early work by Sridharan [26] explored augmented reinforcement learning for interaction with non-experts. A significant challenge in IRL is the quality of human input; Loftin et al. [18] investigated using implicit human feedback strategies, while Faulkner et al. [11] and Kessler Faulkner and Thomaz [13] developed specific algorithms for learning from inaccurate or imperfect teachers. The field has explored various feedback modalities, including haptics [7] and even bio-signals, such as the use of EEG-based implicit human feedback to accelerate learning [29].

2.2.2 VR for Data Collection and Demonstration

The use of VR as a powerful tool for collecting human data for robot learning is well-established. A primary application has been in Learning from Demonstration (LfD) and robot programming, where methods range from multi-resolution corrective demonstration [21] and learning responsive behavior by imitation [3] to Bayesian fusion of multimodal advice [27]. Christen et al. [6] used motion

capture data from human-human interactions to guide the reward function for training dexterous HRI policies. Mosbach et al. [23] used GPU-based simulation and high-quality demonstrations, often collected in VR, to accelerate manipulation learning. More recently, Dogangun et al. [9] developed RAMPA, a system using Augmented Reality for machine programming by demonstration, highlighting the trend towards immersive technologies for intuitive teaching. This approach is also used in neurorehabilitation, as shown by Lydakis et al. [20], who developed a learning-based agent for home therapy using EMG and avatar-based feedback.

2.2.3 VR for Preference and Feedback

While VR for demonstration is mature, its use for providing evaluative feedback is an emerging area. The most directly related work is by de Heuvel et al. [8], who investigated the impact of VR and 2D interfaces on human feedback in *preference-based* robot learning. Their work focuses on comparing interfaces for offline, comparative feedback. Our project aims to extend this line of inquiry into the domain of real-time, in-the-loop *scalar* feedback, which is a distinct and unexplored challenge.

2.2.4 Avatar and Motion Control in VR

A critical technical challenge is interpreting the sparse sensor data from a VR headset into a meaningful representation of human state. This is an area of intense research in computer vision and graphics. Recent breakthroughs have employed Reinforcement Learning to create physically plausible avatars. Winkler et al. [28] presented QuestSim, an RL framework that takes sparse signals from an HMD and two controllers to simulate full-body motion. This work was advanced by Luo et al. [19], who developed a method for a real-time simulated avatar from head-mounted sensors, a technology crucial for providing the context-aware feedback our system intends to use. Further foundational work includes that of Ye et al. [30] on generating physically realistic full-body motion for VR users.

2.2.5 Other Relevant Works

Other relevant works include the integration of feedback in latent space for skill learning [25], the use of fairness-sensitive policy gradients to reduce bias in robotic assistance [31], and the comparison of feedback motion planning with learning from demonstration [15]. The method of using adversarial motion priors as substitutes for complex reward functions [10] is also relevant for shaping agent behavior without manually engineering rewards.

2.3 Limitation and Deficits in the State of the Art

The current state of the art reveals several significant limitations that our project aims to address:

1. **Expressiveness Gap in IRL:** Existing IRL methods predominantly rely on low-dimensional, often binary feedback mechanisms such as keyboard presses or button clicks [11, 18]. These interfaces fail to capture the rich, nuanced feedback that humans can naturally provide through gestures, gaze, and body language, creating an expressiveness bottleneck in human-robot communication.
2. **Underutilization of VR’s Potential for Real-time Feedback:** While VR has been successfully used for collecting demonstration data [9, 23] and more recently for preference elicitation [8], its potential for capturing real-time, in-the-loop scalar feedback within IRL frameworks remains largely unexplored. The multimodal sensing capabilities of modern VR headsets has not been fully used yet for interactive machine learning.
3. **Lack of Multimodal Signal Fusion for Feedback:** While VR provides a multitude of signals (gaze, pose, voice), no existing work in RL systematically investigates which combinations are most effective for providing feedback or how to fuse them robustly for simple HRI tasks.

This project directly addresses these deficits by developing a novel VR-based IRL framework that uses the expressiveness of immersive interfaces and incorporates multimodal signal fusion for a greeting interaction task.

3 Problem Statement

The deficits identified in the state of the art—the expressiveness gap in IRL, the limited use of VR for real-time feedback, and the unresolved challenge of multimodal signal fusion define the core problem of this work: Human feedback is a powerful way to guide RL agents, but existing methods of collecting it are unnatural and provide only a narrow feedback channel (low-bandwidth), and cannot scale to match the pace of modern deep RL training.

We aim to develop a proof-of-concept IRL system that utilizes intuitive, multimodal feedback from a VR headset for a human-robot greeting interaction. Our approach will implement a single-phase pipeline: developing a VR interface that allows a human teacher to naturally give feedback to an RL agent within a simulated greeting task using a combination of inputs (e.g., controller buttons, gestures, gaze direction, and head pose).

The success of this approach will be evaluated by answering the following core research questions:

1. Feasibility: Can multimodal feedback from a VR headset be effectively translated into a robust and informative reward signal for an RL agent in a greeting interaction task? This question targets the deficits of expressiveness and multimodal fusion.
2. Performance Comparison: How does the performance and learning efficiency of an agent trained with rich VR feedback compare to one trained with traditional, low-dimensional feedback (e.g., keyboard inputs)?
3. User Experience: What are the qualitative differences in the teaching experience between VR and keyboard interfaces for providing IRL feedback?

The key challenges will be the noise and ambiguity of VR signals, and the technical complexity of building a functional pipeline between the VR application and the RL training framework.

4 Greeting Interaction Scenario

4.1 Concept

The core concept of the interaction scenario is for a robot to learn to select the correct greeting behavior based on the cues presented by an approaching human. The desired social mapping is straightforward: if the human avatar extends a hand, the correct response is a handshake; if the human is in a neutral stance at a far distance, the robot should wave; and if the human is standing still in close proximity, a bow is the expected greeting. While a similar greeting task has been explored using multimodal deep reinforcement learning [?], their approach relied on a pre-defined, autonomous reward function. Our work is distinct in its focus on Interactive Reinforcement Learning (IRL), where a human teacher actively provides real-time feedback through a VR interface to shape the robot’s policy, rather than relying solely on a pre-programmed reward signal.

4.2 Environment Setup in Unity

The environment will be constructed within the Unity game engine. The virtual arena will consist of a small room or an open space. A human avatar will be programmed to approach the robot, presenting one of three distinct cues to signal its intent: having a hand extended to request a handshake, maintaining a neutral stance while at a far distance to prompt a wave, or standing still when near the robot to create an expectation of a bow. The robot avatar will be a humanoid model rigged with the three necessary greeting animations: waving, performing a handshake, and bowing.

4.3 State Space

The state space for the reinforcement learning agent is designed to capture the essential elements of the social context. It consists of the human’s distance from the robot, which will be discretized into three categories: near, medium, and far. It also includes the human’s current gesture cue, which can be hand extended, neutral, or standing still. Optionally, the agent’s previous action may be included in the state to provide a minimal memory of the interaction history.

4.4 Action Space

The agent operates in a discrete action space comprising four possible actions. It can choose to execute one of the three socially appropriate greetings—wave, handshake, or bow—or it can elect to do nothing.

4.5 Reward Functions

Three distinct reward mechanisms will be implemented and compared. For the normal, autonomous Reinforcement Learning baseline, a pre-defined reward function will be used: the agent receives a +10 reward for performing the correct greeting, a -5 penalty for an incorrect greeting, and a small -1 penalty for every timestep of delay to encourage timely responses. In the Interactive Reinforcement Learning condition with keyboard feedback, the human teacher provides a scalar reward signal by pressing a '+' key for correct behavior and a '-' key for incorrect behavior after each greeting attempt. The final condition, Interactive RL with VR multimodal feedback, aims to utilize more natural human input. This includes explicit controller input for positive or negative feedback, implicit feedback derived from head nod or shake detection, and, as a stretch goal, voice cues such as saying "yes" or "no".

4.6 Evaluation Metrics

We will measure the system's performance using both numbers and user feedback. The main number-based metrics will be the accuracy of the robot's greetings and how many training sessions it takes to reach 80% accuracy. We will also use other measures like the total reward the agent collects and a confusion matrix (with its F1 score) to see exactly where the robot gets confused, for example, if it tries to shake hands when it should bow. We will directly compare the learning progress of the three methods: standard RL, keyboard feedback, and VR feedback. Finally, we will collect user opinions to understand the difference between teaching with a keyboard and teaching in VR.

5 Project Plan

5.1 Work Packages

The project plan will include the following packages:

WP1 : Literature Review and Technical Specification: This initial package involves a review of IRL algorithms, VR development with Unity ML-Agents, and real-time data pipelining. The output will be a detailed technical specification document outlining the chosen RL algorithm, the architecture for the VR-RL communication, and the specific multimodal signals to be captured.

WP2 : Development of the VR-IRL Training Environment: This is a core technical package focused on implementation. It is split into two parallel tracks:

- WP2.1: Greeting Task Development: Creation of the greeting interaction scenario in Unity with a humanoid robot capable of 3 animations (wave, handshake, bow) and a human avatar with predefined interaction cues.
- WP2.2: VR Feedback Interface and Data Pipeline: Development of the VR user interface for feedback. This includes programming the logic to map specific human actions into scalar reward values and building a communication pipeline between Unity and the Python-based RL training framework.

WP3 : Baseline Implementation and System Integration

- WP3.1: Autonomous RL Baseline: Training a standard RL agent using a predefined reward function to establish a performance baseline without human feedback.
- WP3.2: Keyboard Feedback IRL Baseline: Implementation of a traditional IRL baseline where a human provides discrete feedback via a keyboard.
- WP3.3: VR-IRL System Integration: Final integration and testing of the complete VR-IRL system, ensuring stable operation and data logging.

WP4 : Evaluation and Analysis: This package involves the training runs and analysis to answer the research questions.

- WP4.1: Comparative Training: Conducting training sessions using all three methods (autonomous RL, keyboard IRL, VR IRL) with the project member as the human teacher.
- WP4.2: Performance Evaluation: Quantitative evaluation and comparison of all trained agents. Metrics will include learning curves, final task performance, and convergence time.
- WP4.3: Qualitative Analysis: Assessment of the user experience and practical differences between the keyboard and VR teaching interfaces.

WP5 : Project Report and Documentation: Compilation of all results and discussion into the final R&D report. Preparing the codebase for release with documentation.

5.2 Milestones

- M1 : Literature review completed and project framework finalized. Detailed technical plan, including selected algorithms and task design. Completion of WP1.
- M2 : Greeting task environment and VR interface developed and operational. Successful demonstration of the basic greeting interaction in Unity. Completion of WP2.
- M3 : All training systems implemented and baselines established. Autonomous RL and keyboard IRL agents trained, and VR system fully integrated. Completion of WP3.
- M4 : Comprehensive evaluation completed. All comparative training and analysis finished, research questions addressed. Completion of WP4.
- M5 : Final report submitted and code published. Completion of WP5.

5.3 Project Schedule

The project will follow a 6-month timeline. The schedule allocates time for the key development phases with appropriate buffers for troubleshooting and iteration.

The Gantt chart below visualizes the schedule.

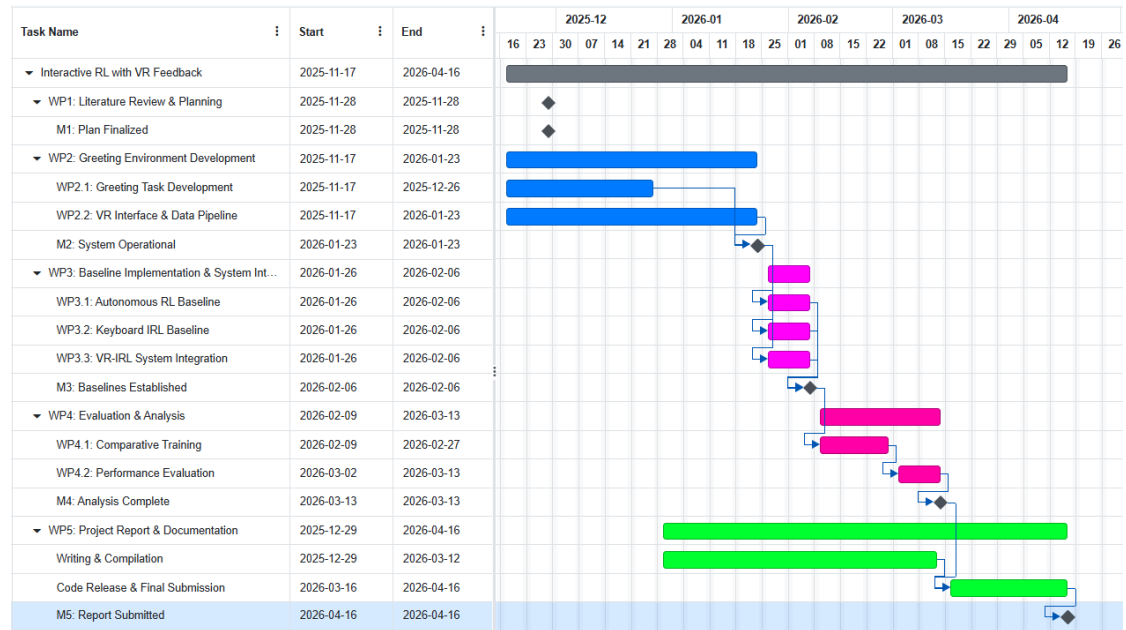


Figure 1: Project schedule Gantt chart, showing the planned duration and overlap of Work Packages (WP1-WP5) and the key Milestones (M1-M5). The OnlineGantt web application (www.onlinegantt.com) was used to create this Chart

5.4 Deliverables

Minimum Viable

- A functional Unity environment for the greeting interaction task.
- A basic keyboard-based IRL interface.
- A trained autonomous RL baseline agent.
- A final report documenting the implemented system and baseline results.

Expected

- All Minimum Viable deliverables, plus:
- A fully operational VR-based feedback system for the greeting task.

- A trained keyboard IRL agent and a VR IRL agent.
- Quantitative comparison showing performance differences between autonomous RL, keyboard IRL, and VR IRL.
- Qualitative analysis of the user experience differences between interfaces.
- A final report with clear analysis and conclusions.

Desired / Future Work

- All Expected deliverables, plus:
- Implementation of multiple feedback modalities in VR (beyond basic controller input).
- Extended evaluation with additional participants.
- A well-documented, open-sourced code repository to support future research.
- Exploration of a predictive feedback model (original desired goal).

Concrete Project Deliverables

The following is a list of the concrete deliverables that will be produced as part of this project:

- D1: Literature survey report. (From WP1)
- D2: A working Unity VR application for IRL feedback collection. (From WP2)
- D3: Baseline performance results and trained agents. (From WP3)
- D4: Complete evaluation results comparing all training methods. (From WP4)
- D5: Final project report and code repository. (From WP5)

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